FREELANCE · REMOTE · AWS CERTIFIED · BOOKING NEW PROJECTS

CHINTALA MAHINDRA

From MVP to millions of requests.

Full-Stack & Cloud Engineer System Design · Spring Boot · React · AWS · Docker · Kubernetes · CI/CD

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PROFILE

Full-stack and cloud engineer who designs, builds and scales products end to end: **React and Next.js** front ends, **Java 21 / Spring Boot 3** services, and **AWS** infrastructure running on **Docker and Kubernetes** with automated CI/CD. I start from the system design (caching, queues, idempotency, isolation, failure modes) and write the decisions down before the code, so clients know what they are getting, why, and what it will cost to run.

WHAT I DELIVER

- **Full-stack MVPs** taken from idea to production on AWS, built to grow rather than be rewritten at the first customer.
- **Scale & performance rescues**: load tests that reproduce the pain, query and cache fixes, and a capacity plan for the next 10×.
- **Kubernetes & cloud migrations**: containerised services on EKS via Terraform and Helm, with zero-downtime cutover.
- **CI/CD platforms**: GitHub Actions, image scanning, GitOps with Argo CD, canary releases with automatic rollback.
- **AI features**: permission-aware RAG and document Q&A with citations, cost controls and evals.

EXPERIENCE

Tata Consultancy Services

AUG 2024 – PRESENT

Software Engineer, Java Backend · Hyderabad · Enterprise HRMS & finance microservices

- Build and enhance **Spring Boot 3 microservices on Java 21** for payroll, workforce and finance workflows, with REST APIs and Spring Data JPA over PostgreSQL and SQL Server.
- Secured employee, HR and admin endpoints with **Spring Security, JWT and role-based access control**.
- Added **asynchronous event processing** and **Redis caching** for high-frequency payroll and attendance lookups.
- Integrated **Spring AI** into existing services for document and policy Q&A, with access control applied to retrieved context.
- Production debugging and API / database issue resolution across backend services in Agile delivery.

SEE IT LIVE

MAIN-HUB-MU.VERCEL.APP

SYSTEMS LAB
Interactive scaling model: traffic, autoscaling, caching, queues and replicas, with the bottleneck explained.

6 ARCHITECTURES
Clickable diagrams with live request traces, scaling stages, decisions and failure modes.

DELIVERY PLATFORM
CI/CD run with canary and automatic rollback; Docker, Kubernetes, Actions and Terraform configs.

EXPERTISE

SYSTEM DESIGN
Scalability, caching, queues, idempotency, multi-tenancy, event-driven design, consistency, capacity planning

BACK END
Java 21, Spring Boot 3, Spring Security (JWT, OAuth2, RBAC), REST, WebSockets, microservices

FRONT END
React, Next.js, TypeScript, Tailwind CSS

DATA
PostgreSQL / Aurora, Redis, DynamoDB, pgvector, SQL Server

AWS
EKS, EC2, Lambda, RDS, S3, CloudFront, SQS, SNS, IAM, Secrets Manager, CloudWatch

PLATFORM
Docker, Kubernetes, Helm, Terraform, Argo CD, GitHub Actions, OpenTelemetry

AI INTEGRATION
Spring AI, RAG, embeddings, vector search, LLM APIs, structured outputs

HOW I WORK

1 · Discovery call
Product, users, pain, deadline.

2 · Design doc & fixed quote
Diagrams, decisions, trade-offs.

3 · Weekly sprints
Working demo on staging every week.

4 · Launch
Load-tested, monitored, rollback rehearsed.

5 · Handover
Runbooks; you own the repo and AWS account.

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SYSTEM DESIGN CASE STUDIES

REFERENCE ARCHITECTURES

Designs I offer clients, each documented with an interactive diagram, request trace, scaling stages, decisions and failure modes on the portfolio.

TENANTFORGE

MULTI-TENANT B2B SAAS

Tenant isolation enforced by Postgres Row-Level Security; pool-then-silo tenancy via a tenant catalog; per-tenant rate limits in Redis; idempotent Stripe webhooks; KEDA-scaled workers on EKS.

SWIFTCART FLASH

FLASH-SALE COMMERCE

Waiting room to shape spikes; atomic Redis inventory with timed reservations so nothing oversells; idempotent checkout and payments; scheduled pre-scaling and k6 rehearsals.

LEDGERLINE

PAYMENTS & LEDGER

Double-entry, append-only journal; serializable transfers; idempotency keys end to end; transactional outbox for events; daily reconciliation and a WORM audit archive.

ROUTELIVE

REAL-TIME TRACKING & DISPATCH

Location stream partitioned by rider; Redis GEO for live dispatch; stateless WebSocket gateways with pub/sub fan-out; PostGIS history; city-level partitioning to scale.

DOCMIND

PERMISSION-AWARE RAG

Permission filter inside the vector query so restricted text never reaches the prompt; cite-or-decline answers; semantic cache; queue-based embedding by content hash; per-team token budgets and evals gating prompt changes in CI.

OPEN SOURCE & PROJECTS

ARGUS — Event-driven industrial IoT monitoring platform

8 SPRINTS

OPEN SOURCE

Java 17 · Spring Boot 3.2 · AWS Lambda · SQS FIFO · SNS · DynamoDB · S3 · Athena · Python · github.com/ARGOD2213/ARGUS

- Always-on **Lambda rule engine** fed by SQS FIFO, publishing CRITICAL / WARNING alerts through SNS; alerts stored in DynamoDB with evidence.
- **Permit-to-Work workflow** with a full state machine, compliance evidence export, shift handover history and nine operator dashboards.
- Named **local predictive ensemble** powering machine chat as advice only, with explicit human approval states.
- Architecture locked by **ADRs**: zero write path to plant equipment, Athena + S3 for history, edge agent with ring buffer and offline rules; no ECS, ALB or NAT to keep cost lean.

PolicyDocs RAG Service

2025 – PRESENT

Spring Boot 3 · Spring AI · Java 21 · PostgreSQL + pgvector · Redis · Docker · AWS

- Document ingestion, chunking, embeddings and **RAG-based Q&A** over policy documents with structured outputs.
- **pgvector** semantic retrieval and conversation context; JWT and RBAC constrain what retrieval can return.
- Containerised with Docker on AWS EC2; grounded answers with no open LLM access.

Digital Library Platform

2025 – PRESENT

Spring Boot 3 · Java 21 · React · PostgreSQL · Redis · S3 · SES/SNS · Docker · AWS

- REST APIs with Spring Data JPA and **Flyway** migrations; role-based workflows for readers, vendors and admins.
- JWT authentication, **Redis caching**, S3 storage for book files, SES/SNS notifications.
- Docker Compose on AWS EC2, deployed by **GitHub Actions CI/CD**.

STANDARDS

Infrastructure as code — Terraform for every AWS resource; separate staging and production.

Delivery — CI on every PR, image and dependency scanning, GitOps with Argo CD, canary or blue-green releases.

Observability — OpenTelemetry traces, Grafana or CloudWatch dashboards, alerts on SLOs.

Security — least-privilege IAM roles, Secrets Manager, OWASP checks, encryption at rest and in transit.

Quality — unit and Testcontainers integration tests, k6 load test before launch, OpenAPI docs.

Ownership — your GitHub and AWS accounts, ADRs, runbooks for every alert.

ENGAGEMENTS

Fixed scope

Clear deliverables, fixed quote, paid by milestone.

Sprint based

Two-week sprints, demo at the end of each.

Retainer

Reserved monthly hours as a fractional backend engineer.

CERTIFICATIONS

AWS Certified Cloud Practitioner

CLF-C02 · 2024

Spring Boot 3 & Spring Framework 6

Udemy (Chad Darby) · 2025

REST API Design, Development & Management

Udemy · 2024

EDUCATION

B.Tech, Electronics & Communication Engineering

VIT Vellore · 2020 – 2024